Task 0 : S.C.I.E.L

Surface Control & Intelligent Exploration Locomotive

Introduction and Overview

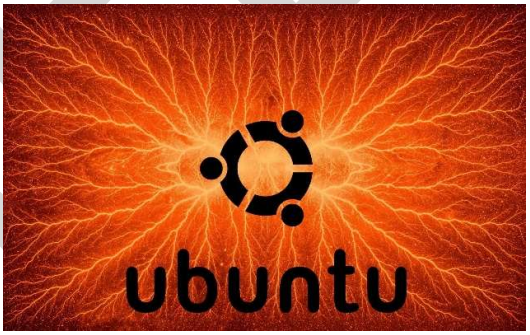
This task aims to familiarize myself with all the given sub-systems included in a rover and to provide a brief understanding of all 14 sub-systems and submit a detailed analysis report of any one sub-system. My report gives an analysis of the traversal sub-system that has not been explicitly stated in the standard regulations, thereby this research presents a conceptual design that builds upon existing approaches that improve mobility and adaptability in terrain.

Brief Analysis of all Sub-systems

Operating System:

An operating system is system software that maintains communication between the software and hardware of a device. There are different types of operating systems, the most popular ones include Windows, MacOS and Linux. Linux is another powerful operating system that is widely used in industrial purposes such as powering smart cars, supercomputers and home appliances. The most stable and open-source Linux Operating System available is Linux Ubuntu. ROS is a type of software that is not exactly an operating system but rather a middleware that sits between a computer's operating system and hardware and helps developers to connect different sensors and motors without having to recode everything from scratch.

S.C.I.E.L will comprise of a Linux-based operating system with ROS integration that will be used for modular communication and efficient handling of the navigation system.



Computing Device

To handle real-time data processing and decision making, the rover requires a computing device. The computing device integrates inputs across subsystems and ensures co-ordinated operation under dynamically changing weather and terrain. S.C.I.E.L will consist of a Raspberry Pi (16GB) to handle navigation and is supported by microcontrollers such as the ESP-32 or STM-32 for low level motion control.

In S.C.I.E.L the Raspberry Pi will act as a decision-making unit that handles path

planning and navigation logic. The microcontroller will also oversee running high-level algorithms for autonomous operations such as well as process camera vision data. The Esp-32 shall focus on handling motor control and reading sensor data to forward the information to the Pi. The ESP will act as a bridge between the Pi and hardware.

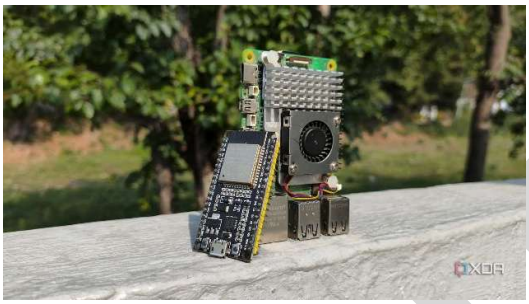


Image source : https://www.xda-developers.com/the-esp32-is-replacing-raspberry-pis-in-my-home-lab-and-it-costs-a-fraction-of-the-price/

Sensors

The sensors allow the rover to perceive it's surroundings by collecting data such as distance, orientation and features of the terrain helping the rover to adapt to changing conditions. Sensors help the rover to interact with its environment safely.

Just like Perseverance, S.C.I.E.L will use a SuperCAM (Image 1) which is a powerful laser that is used for scanning objects up to 9 meters away along with MEDA (Image 2) to read the weather and atmosphere of the Martian surface.

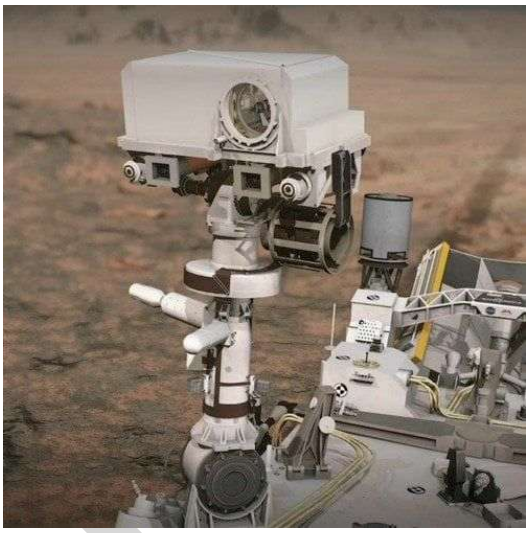


Image source : https://www.reddit.com/r/PerseveranceRover/comments/m245dl/perseverance_audio_recordings_from_mic_on/

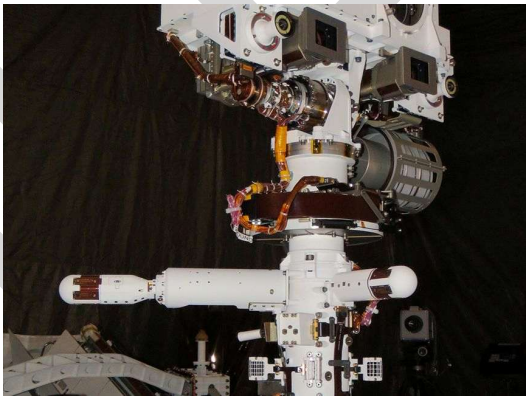


Image source : https://science.nasa.gov/mission/mars-2020-perseverance/science-instruments/#sensors

Mobility & Locomotion

By reading the environment, the rover should be able to process the data and make decisions thereby avoiding obstacles, planning paths and maintaining stability in its adventures.

By processing visual data from the MastCAM-Z and NavCAMS (Also from Perseverance) the data will be processed by the Pi and will assess the terrain and feed the information to the ESP that assists the drivetrain in switching between

wheeled and leg-assisted movement modes.

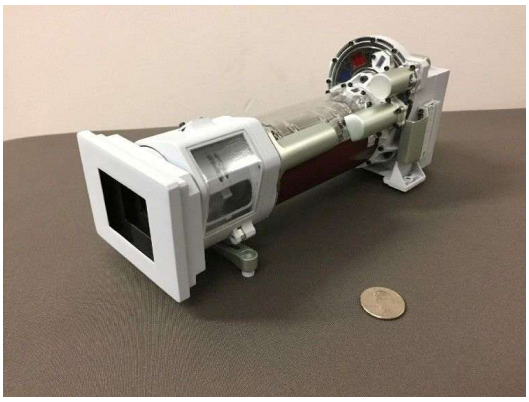


Image source : https://indianexpress.com/article/technology/science/nasa-perseverance-mars-2020-rover-mastcam-z-6394993/

Communication System

The communication system allows the rover to send data to Earth and as well as receive commands. The communication system must be reliable and capable of handling delays and survive environmental interference while transmitting information.

The rover will transmit information to the five NASA satellites that orbit around Mars and relay the information back to Earth.

Control System

The control system coordinates all subsystems and ensures proper execution of tasks. The control subsystem must be able to adapt to unexpected situations and respond to changing mission objectives. The control subsystem can be considered the backbone of the rover that coordinates sensors and navigation. S.C.I.E.L's control system will ensure smooth switching between movement modes.

Power Electronics

The power electronics system manages power generation and distribution. It ensures that all components receive enough power to perform the tasks requirements.

S.C.I.E.L consists of a regulated power distribution that will prioritize critical systems such as drivetrain and control for reliable operation.

Manipulator

The manipulator is a complex system that brings together various mechanical electronics and software components to perform a specific task. It performs tasks such as drilling, collecting samples or placing instruments thereby requiring precision and flexibility.

As of right now S.C.I.E.L's focus is terrain navigation and mapping for future expeditions.

Chassis

The chassis is a framework that can be considered as the main body of the rover. The chassis ensures the mechanical stability of the rover and most importantly house and protect subsystems.

S.C.I.E.L will consist of a lightweight Aluminum alloy chassis with Titanium alloy located in regions of high stress.

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Image source : https://www.jpl.nasa.gov/news/nasas-perseverance-rover-goes-through-trials-by-fire-ice-light-and-sound/

Drivetrain

The drivetrain helps the rover to traverse terrain using wheels and suspension systems. It ensures traction and stability. A hybrid leg-wheel suspension system will be implemented in S.C.I.E.L combining 6 wheels with legs to improve terrain adaptability. A full analysis of terrain adapting drivetrain has been explained below.

Below image attached is the common rocker-bogie drivetrain which is found on both the Curiosity and Perseverance Mars rovers.

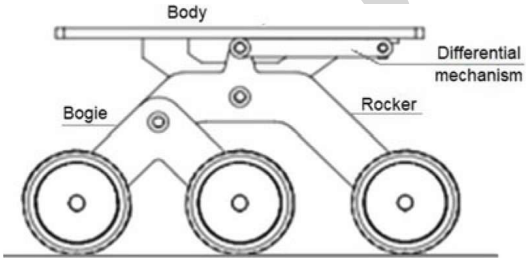

Image source : https://www.cambridge.org/core/journals/robotica/article/theoretical-study-on-a-modified-rockerbogie-suspension-for-robotic-rovers/06411968B7D4807F7E062DF2870FFDEE

Electronics Safety System

This system protects electronic components from damage due to overloading caused by incorrect electrical inputs. It ensures reliability and prevents mission failure due to electrical issues.

Life Detection system

The life detection system is used for identification of potential signs of life from the past or present by analyzing samples that have been collected. This involves chemical and environmental analysis to detect biological indicators.

Science Cache

The science cache contains instruments used to conduct experiments and analyze environmental and geological data of the Martian surface and atmosphere.

Geology of Mars

Mars has the most explorable terrain in our solar system outside of Earth, which sounds like a boastful statement until you realize our other two options are Mercury that is a planet with scalding landscape and Venus being a hellish planet filled with volcanoes, sulphuric ash and acid rain – Mars comparatively seems to be the most peaceful out of the other two neighbors.

The surface of Mars is covered with dust, mainly Ferric Oxide which is found in rust that is responsible for the dominant red orange color of Mars whereas previous

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studies suggested Hematite to be the primary reason for the red hue suspended on the surface. The terrain itself is unforgiving with rocky hills, mountains, craters and volcanoes with winds reaching upto 20m/s, the presence of dust and sand itself pose problems in the mechanical sub-system which result in motor failure due to sand particles clogging joints and axles that lead to inefficient movement caused by increased friction between joints.

The following pages shall introduce my design proposal for mobility and traversal systems of the rover.

Sub-system

Mobility & Locomotion

Before a rover can explore on Mars, it first must survive moving on it. This makes mobility and structural design one of the most crucial parts for creating a Mars rover.

Traditionally Mars rovers use wheels to traverse the rough and coarse terrain of the Martian surface. While this mode of traversal is energy efficient and reliable for a plethora of reasons, it limits the mobility of the rover in irregular terrain thus restricting exploration.

The Proposal:

To address this limitation, my proposed solution is a design that takes inspiration from the AT-TE from Star Wars (Refer to the picture below). The proposed system is a hybrid 6 wheeled and legged transport mechanism where the rover can switch its mode of traversal depending on the terrain conditions, thus making the rover a versatile exploration expeditioner on Martian territory.

(Image source: https://starwars.fandom.com/wiki/All_Terrain_Tactical_Enforcer/Legends)

Reasons and Advantages:

- **Improved Performance on loose soil:** Martian soil is loosely packed and very fine in terms of texture; however this increases the odds of slippage from wheels or the rover getting stuck or being sunken in the sand. Walking movement reduces the reliance on traction by actively lifting and placing the rover thus minimizing the chance of wheel spin.

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- **Enhanced Climbing:** Wheeled rovers are limited by sloped angles and traction of the wheels. Leg based mechanisms can maintain stability on steeper inclines by adjusting its posture and having a better grip to climb more easily.
- **Stability on uneven surfaces and in high-speed winds:** A multi-legged configuration allows for balancing the rover in extreme conditions such as strong winds carrying dust across terrain. This reduces the risk of tipping over compared to wheel systems especially when travelling highly uneven terrain.
- **Redundancy and fault tolerance:** In multi-legged systems, the failure of one limb does not jeopardize the mobility of the rover. The system can be reconfigured to compensate the remaining legs and thus increasing mission reliability.

Selection of Materials

In order to design a rover we need to select the best materials at our disposal that can withstand the harsh conditions of Mars, which include strong winds, dust storms, extreme temperature gradients, abrasive dust, extremely low atmospheric pressure (vulnerability to cosmic rays and magnetic disturbances) and vast terrain which undergo immense altitude and texture changes. Below consists of a breakdown of the type of materials required for the chassis, wheels, legs and joints.

1. **Main Body / Chassis:**

   My proposed two options for the standard integrity while being very lightweight compared to Titanium. Additionally Aluminum is very resistant to corrosion and can manage temperature variations.

   Titanium alloys can be used in the regions undergoing high stress due to its nature of having superior strength and resistance to fatigue as well as deformation, however this comes at a large commercial cost as Titanium is very expensive to obtain.

2. **Legs:**

   The primary and sole choice for this part is Titanium. The legs experience repeated shock and fatigue especially when travelling in rough terrain. Titanium is one of the best options due to its high strength and lower weight not to mention it's excellent fatigue resistance thus making it suitable for long duration missions.

3. **Joints:**

   The main concern regarding Martian surface is that it is filled with dust which increases the chances of corrosion and thus making the joints more prone to

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failure and clogging. Joints undergo continuous rotation and stress; they must resist wearing down and especially be resistant to dust which means minimum to zero exposure. Traditional liquid lubricants cannot be used be used as Mars undergoes extreme temperature changes as Mars undergoes extreme temperature changes which could freeze the lubricants further locking joints into place.

To combat this problem, we may stick to Titanium or hardened and reinforced steel paired with dry lubricants such as Graphite or MoS_2.

4. Wheels :

The wheels need to be sturdy enough to sustain the weight of the rover (~1200 kg / 2640 lb). The idea is to use thick Aluminum wheels like Perseverance Mars Rover with a much broader base.

The thickness is mainly due to avoiding cracks and tears that crippled Curiosity Mars Rover as it had a thin base for its Aluminum wheels.

Attached below are images of the rover wheels for reference.

(Image source : https://www.space.com/mars-rover-perseverance-wheels-parachutes.html)

5. Footpads :

The footpads of the legs can be made wider in a circular design as well as textured to absorb and distribute load and improve grip on the soil thereby reducing the risk of sinking. The base of the footpad can be made with Titanium alloys because of all the constant load bearing nature of the parts and to again avoid punctures and tearing of parts.

Disadvantages of leg-wheel locomotion

Despite improvements in adaptability to environments and terrain handling, the proposal of a hybrid leg-wheel system poses noteworthy challenges to be overcome. When we consider the current situation in Mars missions, the

disadvantages can outweigh the advantages when we prioritize reliability and energy optimization.

The disadvantages are as follows.

1. **High Energy Consumption:**
   Legged movement requires multiple joints working together in perfect co-ordination and precision making it significantly more power hungry compared to wheel systems. On Mars, where solar panels can be rendered obsolete due to dust and power be limited at nights or dust storms. RTG (Radioisotope Thermoelectric Generator) is very powerful when it comes to powering the rover but most of the energy can be instead utilize in locomotion rather than sample analysis and communication.

2. **Mechanical Complexity:**
   The increase in number of legs and joints increase the number of failure points. Especially in an area where repair is either limited or unavailable, this reduces system reliability.

3. **Control Complexity:**
   Algorithms responsible for control and co-ordination of multiple legs require precise control and synchronization. By increasing the complexity of onboard systems, we increase the chances of facing errors during operation.

4. **Higher mass:**
   Inclusion of additional structures and legs increases the rover's mass which can directly impact problems with fitting the rover in a rocket and increase the launching costs of the rover while also increasing the energy usage of the rover.

5. **Wear due to dust:**
   The environment of the Martian Surface has fine dust that can quickly accumulate in joints and moving parts which can lead to wear and potentially malfunctioning in a short period of time. The dust, sand and dirt could cause clogging issues and thus cripple the rover.

These limitations highlight why simple wheel systems can outperform legged walkers due to greater reliability. While the hybrid approach offers better adaptability, it may be more suited for future missions where advancements in energy systems and control systems can support its complexity.

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Working Mechanism

Before SCIEL moves, it doesn't just "drive forward" like a normal vehicle. It first tries to understand the ground beneath it.

Using its cameras and onboard sensors, SCIEL constantly observes the terrain ahead. It looks for changes in slope, loose soil, rocks, and gaps. Based on this input, the rover decides how it should move rather than blindly moving forward.

Normal Movement – Wheeled Mode

On relatively flat and stable terrain, S.C.I.E.L behaves like a conventional rover.

All six wheels remain in contact with the ground, and each wheel is powered independently. This allows smooth forward motion while also enabling slight adjustments in direction. Since wheels consume less power compared to legs, this mode is used as much as possible to conserve energy.

Adapting to terrain – Legged Mode

As SCIEL encounters more difficult terrain, such as loose soil or uneven rocks, it begins to change its behaviour.

Sensors in the wheels and footpads detect reduced traction or instability. At the same time, the cameras identify obstacles ahead. Instead of forcing its way through, SCIEL slows down and prepares to adapt.

In this mode, the wheels temporarily **lock their rotation**, preventing slipping. Each wheel assembly then rotates approximately 90 degrees inward, changing its orientation from a rolling surface to a stable contact surface. Once repositioned, these wheels effectively function as footpads for the legs.

With the wheels now acting as feet, the legs take over the primary role in movement. Each leg lifts, extends, and places the "foot" carefully onto stable ground. This allows the rover to step over obstacles, maintain balance on uneven surfaces, and avoid sinking into loose terrain.

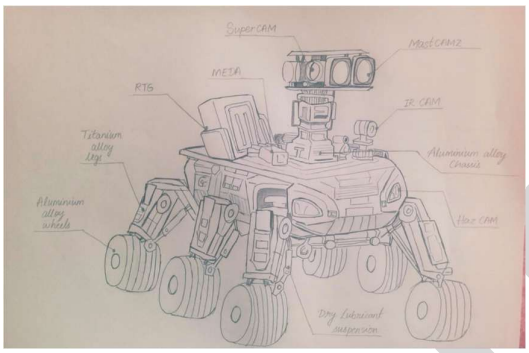
Maintaining Stability and Balance

One of the most important aspects of SCIEL's movement is balance.

At any given moment, multiple legs and wheels remain in contact with the ground. The rover continuously adjusts the position of each leg to keep its centre of mass stable. If one side of the rover encounters a higher obstacle, the legs on that side extend while others compensate to maintain equilibrium. This constant adjustment happens in real time, making the rover appear almost "aware" of how it is standing. With the wheels now acting as feet, the legs take over the primary role in movement. Each leg lifts, extends, and places the "foot" carefully onto stable ground. This allows the rover to step over

obstacles, maintain balance on uneven surfaces, and avoid sinking into loose terrain.

The transformation mechanism is designed to ensure that the contact surface of the wheel is repositioned to maximize stability and load distribution when acting as a footpad.



The above image is a diagrammatic representation of S.C.I.E.L that has been hand-drawn to illustrate the robot's appearance. The dimensions of the robot are listed below in an approximate estimate.

- Length**:** 2.5 – 3 meters
- **Width:** 2 – 2.5 meters
- **Height:** 1.8 – 2.2 meters (can extend slightly in leg mode)
- Weight**:** 250 – 400 kg (scaled-down prototype class)

Conclusion

The hybrid design of SCIEL promotes adaptability to many types of Mars' surface terrain, using both legs and wheels for movement using hybrid means of propulsion. Since SCIEL can utilize fuel efficient motion and leg assisted means of locomotion, it will provide more traction than a traditionally designed wheeled vehicle, thus enabling it to effectively traverse all terrain types within Mars's environment. While able to lock and re-position their wheels into footpads, the vehicle will be able to lock into position during use in difficult terrain and assist in providing better control of motion and stability when travelling through rough terrain.

While there are many disadvantages of using such a system within SCIEL's design, the vehicle will have increased complexity with regards to the mechanical systems that are needed to operate the legs and wheels, which will increase energy consumptions and require additional human or autonomous control for the operation of the vehicle. Hence, given the current operational efficiency and reliability of existing rovers, one can say that the disadvantages of the hybrid vehicle will outweigh the advantages that this design provides based on the configuration of the current status Mars Exploration Programs.

Regardless of the compromises created by this design, it offers a new and different way to explore planets such as Mars. With further advances in materials, energy sources/propulsion systems, and control algorithms, further hybrid designs will create ways for us to transport to places on

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Mars where we otherwise would not have access. Thus, S.C.I.E.L's design indicates a balance between current and future use and demonstrates how innovative mechanical design can expand our capabilities for exploring other planets.

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